Jackson Chen

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EDUCATION

Princeton University Princeton, NJ
Intended A.B. of Computer Science Expected May 2030

- Intended Minors: Statistics and Machine Learning, Cognitive Science
- Activities: Princeton Racing Electric, Entrepreneur Club

TECHNICAL EXPERIENCE

Hoagie.io Princeton, NJ
Software Developer, Full-Stack September 2026 – Present

- Developed features for and maintained HoagiePlan, a Princeton student planning platform used by 1,000+ students, with Next.js, React, TypeScript, Django, and PostgreSQL.
- Collaborated with student developers through Git-based development workflows to design, implement, test, and deploy new application functionality.

NYC Spark Lab New York, NY
Research Intern, Project Lead May 2025 – June 2026

- Led computational analysis of hazardous heat exposure across 500+ Native American tribal regions, developing reproducible geospatial workflows in R to process population and environmental datasets.
- Redesigned Spark Lab's wiki to centralize technical documentation, improving accessibility for professors, researchers, and interns across lab projects and resources.

NYU mLab New York, NY
Machine Learning Researcher, Volunteer April 2025 – July 2025

- Annotated network traffic data and mapped domain activity to iOS applications, improving training data quality for RouterSense.AI's behavioral analysis model.
- Tested and evaluated machine learning models within a data pipeline for network-based behavioral analysis.

TECHNICAL PROJECTS

Hazardous Heat Exposure in Native American Indigenous Territories May 2025 – June 2026

- Processed 500+ Native American tribal regions and 21 years of WorldPop population rasters in R to quantify population exposure to hazardous heat across tribal lands.
- Built reproducible data-preparation workflows to load, clean, summarize, and aggregate WBGT and population data at tribal and state levels, including multi-state territories and geographic boundary standardization.
- Automated raster extraction and validation, generating yearly population summaries from 2000–2020 gridded datasets.

Equitable Access to Urban Heat Adaptation Infrastructure in NYC November 2025 – April 2026

- Built an end-to-end Python geospatial pipeline across 8 sequential modules for preprocessing, data import, WBGT calculation, adaptive-capacity analysis, visualization, and disparity statistics.
- Developed a Composite Heat Vulnerability Index integrating WBGT heat exposure, CDC Social Vulnerability Index data, and access to cooling centers, green space, and urban-design infrastructure.
- Integrated Census tract, NTA, community district, PRISM climate, historical redlining, and infrastructure datasets to analyze relationships between historical housing discrimination and present-day heat vulnerability.

Construction Workers On the Go (COGO) July 2024 – August 2024

- Led development of a C++ wearabledo safety system integrating Arduino, biosensors, environmental sensors, OLED displays, and real-time hazard alerts for construction workers; earned 1st place at NYU Tandon IESOSC.
- Implemented ultrasonic proximity detection, Bluetooth communication, noise monitoring, and moving-average light sensing across smartwatch and helmet modules.

ADDITIONAL

Technical Skills: Python, R, C++, Java, TypeScript, SQL, Git, GitHub.
Frameworks and Tools: React, Next.js, Django, PostgreSQL, OpenCV, ArcGIS.
Machine Learning: Regression, Decision Trees, Naive Bayes, Clustering, Computer Vision.